# SHANXI KNLAN TECH CO., LTD.

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## MATERIAL SAFETY DATA SHEET

## Calcium Nitrate Tetrahydrate

### 1. IDENTIFICATION OF THE MATERIAL AND SUPPLIER

Product Name: CALCIUM NITRATE TETRAHYDRATE

Other name(s): Calcium nitrate; Calcium nitrate tech; Calcium Nitrate - Fertiliser Grade.

Recommended Use: Fertilizer, raw material.

Supplier: SHANXI KNLAN TECH CO.,LTD.

### 2. HAZARDS IDENTIFICATION

NON-HAZARDOUS SUBSTANCE.

### 3. COMPOSITION/INFORMATION ON INGREDIENTS

Components:Total Nitrogen 11.7%,Calcium Oxide 23.4%

### 4. FIRST AID MEASURES

Inhalation: Remove victim from area of exposure - avoid becoming a casualty. Remove contaminated clothing and loosen remaining clothing. Allow patient to assume most comfortable position and keep warm. Keep at rest until fully recovered. If patient finds breathing difficult and develops a bluish discolouration of the skin (which suggests a lack of oxygen in the blood - cyanosis), ensure airways are clear of any obstruction and have a qualified person give oxygen through a face mask. Apply artificial respiration if patient is not breathing. Seek immediate medical advice.

Skin Contact: If skin contact occurs, remove contaminated clothing and wash skin with running water. If irritation occurs seek medical advice. Nitrates can be absorbed through cut, burnt or broken skin. Launder contaminated clothing before reuse.

Eye Contact: If in eyes, wash out immediately with water. In all cases of eye contamination it is a sensible precaution to seek medical advice. Ingestion: Rinse mouth with water. If swallowed, do NOT induce vomiting. Give a glass of water. Seek medical advice.

Medical attention and special treatment: Treat symptomatically. Treat as for exposure to nitrates. May cause methemoglobinemia. Clinical findings: The smooth muscle relaxant effect of nitrate salts may lead to headache, dizziness and marked hypotension. Cyanosis is clinically detectable when approximately 15% of the haemoglobin has been converted to methaemoglobin (ie. ferric iron). Symptoms such as headache, dizziness, weakness and dyspnoea occur when methaemoglobin concentrations are 30% to 40%; at levels of about 60%, stupor, convulsions, coma and respiratory paralysis occur and the blood is a chocolate brown colour. At higher levels death may result. Spectrophotometric analysis can determine the presense and concentration of methaemoglobin in blood.

Treatment: 1. Give 100% oxygen. 2. In cases of (a) ingestion: use gastric lavage, (b) contamination of skin (unburnt or burnt): continue washing to remove salts. 3. Observe blood pressure and treat hypotension if necessary. 4. When methaemoglobin concentrations exceed 40% or when symptoms are present, give methylene blue 1 to 2 mg/kg body weight in a 1%

solution by slow intravenous injection. If cyanosis has not resolved within one hour a second dose of 2 mg/kg body weight may be given. The total dose should not exceed 7 mg/kg body weight as unwanted effects such as dyspnoea, chest pain, vomiting, diarrhoea, mental confusion and cyanosis may occur. Without treatment methaemoglobin levels of 20-30% revert to normal within 3 days. 5. Bed rest is required for methaemoglobin levels in excess of 40%. 6. Continue to monitor and give oxygen for at least two hours after treatment with methylene blue. 7. Consider transfer to centre where haemoperfusion can be performed to remove the nitrates from the blood if the condition of the patient is unstable. 8. Following inhalation of oxides of nitrogen the patient should be observed in hospital for 24 hours for delayed onset of pulmonary oedema. Further observation for 2-3 weeks may be required to detect the onset of the inflammatory changes of bronchiolitis fibrosa obliterans.

**5. FIRE FIGHTING MEASURES**

Hazards from combustion products: Non combustible, but will support combustion of other materials. Precautions for fire fighters and special protective equipment: Not combustible, however will support the combustion of other materials. Increases intensity of a fire. Decomposes on heating emitting toxic fumes, including those of oxides of nitrogen . Fire fighters to wear self-contained breathing apparatus and suitable protective clothing if risk of exposure to products of decomposition. Keep containers cool with water spray. Suitable Extinguishing Media: Not combustible, however, if material is involved in a fire use: Water jets. Water spray (large quantities). Unsuitable Extinguishing Media: Dry powder. Carbon dioxide. Halogens.

**6. ACCIDENTAL RELEASE MEASURES**

Emergency procedures: Clear area of all unprotected personnel. In the case of a transport accident notify the Police. If contamination of sewers or waterways has occurred advise local emergency services.

Methods and materials for containment and clean up: Slippery when spilt. Avoid accidents, clean up immediately. Wear protective equipment to prevent skin and eye contact and breathing in dust. Work up wind or increase ventilation. Cover with damp absorbent (inert material, sand or soil). Sweep or vacuum up, but avoid generating dust. Collect and seal in properly labelled containers or drums for disposal.

**7. HANDLING AND STORAGE**

Conditions for safe storage: Store in a cool, dry, well ventilated place and out of direct sunlight. Store away from incompatible materials described in Section.Keep containers closed when not in use - check regularly for spills. Precautions for safe handling: Avoid skin and eye contact and breathing in dust. Avoid handling which leads to dust formation.

**8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

These Exposure Standards are guides to be used in the control of occupational health hazards. All atmospheric contamination should be kept to as low a level as is workable. These exposure standards should not be used as fine dividing lines between safe and dangerous concentrations of chemicals. They are not a measure of relative toxicity. Engineering controls: Ensure ventilation is adequate to maintain air concentrations below Exposure Standards.

Avoid generating and breathing in dusts. Use with local exhaust ventilation or while wearing dust mask. Keep containers closed when not in use.

**9. PHYSICAL AND CHEMICAL PROPERTIES**

Physical state: Granules Colour: White Odour: Odourless Solubility: Soluble in water. Specific Gravity: 1000-1100 kg/m3 (Bulk density) Relative Vapour Density (air=1): Not available Vapour Pressure (20 °C): Not available Flash Point (°C): Not applicable Flammability Limits (%): Not applicable Autoignition Temperature (°C): Not applicable Melting Point/Range (°C): 94-98 pH: 5-7 (10% aqueous solution)

**10. STABILITY AND REACTIVITY**

Chemical stability: Stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

Conditions to avoid: Avoid exposure to heat, sources of ignition, and open flame. Avoid dust generation.

Incompatible materials: Incompatible with combustible materials , reducing agents , acids , metal powders , ammonia , hydrazine , and organic materials .

Hazardous decomposition products: Oxides of nitrogen. Hazardous reactions: Hazardous polymerisation will not occur.

**11. TOXICOLOGICAL INFORMATION**

No adverse health effects expected if the product is handled in accordance with this Safety Data Sheet and the product label. Symptoms or effects that may arise if the product is mishandled and overexposure occurs are: Ingestion: Swallowing can result in nausea, vomiting, diarrhoea, and abdominal pain. Swallowing large amounts may result in headaches, dizziness and a reduction in blood pressure (hypotension). Eye contact: May be an eye irritant. Exposure to the dust may cause discomfort due to particulate nature. May cause physical irritation to the eyes. Skin contact: Contact with skin may result in irritation. Can be absorbed through cut, broken, or burnt skin with resultant adverse effects. Inhalation: Breathing in dust may result in respiratory irritation. Absorption of nitrates by inhalation, ingestion or through burnt or broken skin may cause dilation of blood vessels by direct smooth muscle relaxation and may also cause methaemoglobinaemia. Long Term Effects: No information available for the product. Toxicological Data: Oral LD50 (rat): 3900 mg/kg SKIN: Mild irritant (rabbit). EYES: Mild irritant (rabbit). Following the ingestion of nitrates in humans and animals methaemoglobinaemia has occured.

**12. ECOLOGICAL INFORMATION**

Ecotoxicity Avoid contaminating waterways.

**13. DISPOSAL CONSIDERATIONS**

Disposal methods: Refer to Waste Management Authority. Dispose of material through a licensed waste contractor. Empty containers must be decontaminated by rinsing thoroughly with water. Rinsing water needs to be disposed of carefully.

**14. TRANSPORT INFORMATION**

Road and Rail Transport Not classified as Dangerous Goods

Marine Transport Not classified as Dangerous Goods

Air Transport Not classified as Dangerous Goods

**15. REGULATORY INFORMATION**

Chemicals management regulation(published by State Council)have specific rule on safe storage and delivery.

**16. Other Information**

Warranty

This information herein given in good faith and to the best of our knowledge. Since we have no control over the use of the product, we make no warrantee or representation, expressed or implied, other than that it conforms to the chemical description on its label. No agent, representative or employee of Shanxi Knlan Teh Co.,Ltd. is authorized to change any terms of this notice.